

Math 240 Fall 2023 - Homework 6 Solution

- (4.2.) 44. Define a linear transformation $T : \mathbb{P}_2 \rightarrow \mathbb{R}^2$ by $T(\mathbf{p}) = \begin{bmatrix} \mathbf{p}(0) \\ \mathbf{p}'(0) \end{bmatrix}$. Find polynomials $\mathbf{p}_1$ and $\mathbf{p}_2$ in $\mathcal{P}_2$ that span the kernel of T , and describe the range of T .

Sol. The kernel of T is the set of all polynomials $\mathbf{p} \in \mathbb{P}_2$ such that $T(\mathbf{p}) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. (Note that any polynomial in $\mathbb{P}_2$ with a zero constant term is in the kernel, and all polynomials in the kernel should have this form.) Thus,

$$\ker(T) = \{\mathbf{p} \in \mathbb{P}_2 : \mathbf{p}(x) = a_1x + a_2x^2 \text{ where } a_1, a_2 \in \mathbb{R}\}.$$

Notice that this kernel is exactly the linear combinations of the vectors x and x^2 . We can take $\mathbf{p}_1(x) = x$ and $\mathbf{p}_2(x) = x^2$ to see that they span the kernel of T .

The range of T is the collection of $\begin{bmatrix} \mathbf{p}(0) \\ \mathbf{p}'(0) \end{bmatrix}$ for all $\mathbf{p}$ in the domain $\mathbb{P}_2$. (Each $\mathbf{p}(0)$ is an element of $\mathbb{R}$ and $\mathbf{p}'(0)$ is the constant term of $\mathbf{p}$ while the constant term can be taken freely from $\mathbb{R}$.) Thus the range of T is given by

$$\text{range}(T) = \left\{ \begin{bmatrix} a_0 \\ a_0 \end{bmatrix} \in \mathbb{R}^2 : a_0 \in \mathbb{R} \right\} = \text{Span} \left(\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\} \right),$$

which is a line in $\mathbb{R}^2$.

- (4.3.) 13. Assume that A is row equivalent to B . Find bases for Nul A , Col A , and Row A .

$$A = \begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 & 6 & 5 \\ 0 & 2 & 5 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Sol. The general solution of $Ax = 0$ or equivalently that of $Bx = 0$ is given by

$$x = x_3 \begin{bmatrix} -6 \\ -\frac{5}{2} \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -5 \\ -\frac{3}{2} \\ 0 \\ 1 \end{bmatrix} \text{ and thus we can choose a basis for Nul } A \text{ by } \left\{ \begin{bmatrix} -6 \\ -\frac{5}{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ -\frac{3}{2} \\ 0 \\ 1 \end{bmatrix} \right\}.$$

(Take the first and second columns from A .) The set

$$\left\{ \begin{bmatrix} -2 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} 4 \\ -6 \\ 8 \end{bmatrix} \right\}$$

is a basis of Col A .

Either $\{[-2 \ 4 \ -2 \ -4], [2 \ -6 \ -3 \ 1]\}$ or $\{[1 \ 0 \ 6 \ 5], [0 \ 2 \ 5 \ 3]\}$ (the set of pivot rows from A or from B) is a basis of Row A .

- (4.4.) 13. The set $\mathcal{B} = \{1 + t^2, t + t^2, 1 + 2t + t^2\}$ is a basis for $\mathbb{P}_2$. Find the coordinate vector of $\mathbf{p}(t) = 1 + 4t + 7t^2$ relative to $\mathcal{B}$.

Sol. We wish to find the linear combination of the basis vectors in $\mathcal{B}$ that gives $\mathbf{p}$. Considering the three coefficients of $1, t, t^2$ respectively, this can be written as the following system of equations,

$$\begin{array}{rclcl} c_1 & & + & c_3 & = 1 \\ & c_2 & + & 2c_3 & = 4 \\ c_1 & + & c_2 & + & c_3 & = 7 \end{array}$$

Convert it to matrix system and solve for the vector $[c_1 \ c_2 \ c_3]^T$ which is precisely the coordinate vector $[\mathbf{p}]_{\mathcal{B}}$.

Another way of solving it is using the change of basis technique from 4.6. First read the given basis vectors in the standard basis $\{1, t, t^2\}$ and place them as the columns of a matrix to create the change of basis matrix from $\mathcal{B}$ to the standard basis $\mathcal{S}$.

$$P_{S \leftarrow \mathcal{B}} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}.$$

The coordinate vector of $\mathbf{p}$ then can be obtained by solving the matrix equation

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} [\mathbf{p}]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix}$$

$$\text{by } [\mathbf{p}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 6 \\ -1 \end{bmatrix}.$$

36. Let $\mathbf{p}_1(t) = 1 + t^2$, $\mathbf{p}_2(t) = t - 3t^2$, $\mathbf{p}_3(t) = 1 + t - 3t^2$.

a. Use coordinate vectors to show that these polynomials form a basis for $\mathbb{P}_2$.

b. Consider the basis $\mathcal{B} = \{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ for $\mathbb{P}_2$. Find $\mathbf{q}$ in $\mathbb{P}_2$, given that $[\mathbf{q}]_{\mathcal{B}} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$.

Sol. a. The coordinate vectors of $\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3$ are given by $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ -3 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix}$ respectively.

To study the linear independence, row reduce the matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & -3 & -3 \end{bmatrix} \sim I_3$ to see that

all columns are pivot columns. Thus the set $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ is linearly independent. Since the space $\mathbb{P}_2$ has a basis $\{1, t, t^2\}$ of size 3, any linearly independent vector set of size 3 in this space is a basis. This concludes the proof.

b. Given the coordinate of $\mathbf{q}$ with respect to the basis $\mathcal{B}$, we know that

$$\mathbf{q} = -1\mathbf{p}_1 + 1\mathbf{p}_2 + 2\mathbf{p}_3 = 1 + 3t - 10t^2.$$